NLPrompt: Noise-Label Prompt Learning for Vision-Language Models

Anonymous Author

Abstract

Vision-language models (VLMs) such as CLIP have shown remarkable capability for efficient adaptation via prompt learning. However, existing prompt learning methods train with cross-entropy (CE) loss, which is notoriously susceptible to label noise commonly present in real-world datasets, leading to severe performance degradation especially in the few-shot regime. In this paper, we propose ***NLPrompt****, a noise-robust prompt learning framework for VLMs. We first demonstrate that replacing CE with mean absolute error (MAE) loss in prompt tuning, which we term* ***PromptMAE****, substantially improves robustness to noisy labels. We provide a theoretical justification via feature learning theory showing that MAE increases the effective signal-to-noise ratio of learned representations by attenuating the gradient influence of mislabeled samples. Building on this insight, we further introduce* ***PromptOT****, which leverages optimal transport (OT) with CLIP text-feature prototypes as class anchors to partition training data into clean and noisy subsets in a globally consistent manner. NLPrompt harmonizes PromptMAE and PromptOT: it applies CE on the OT-identified clean subset and MAE on the noisy subset, exploiting the complementary strengths of both losses. Extensive experiments on seven benchmark datasets under various synthetic noise settings and on the real-world noisy Food101N dataset demonstrate that NLPrompt significantly outperforms state-of-the-art methods. Furthermore, NLPrompt generalizes seamlessly to other prompt-tuning frameworks including VPT, MaPLe, and PromptSRC, consistently boosting their noise robustness.*

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1. Introduction

Large-scale vision-language models (VLMs), exemplified by CLIP [32], have transformed image recognition by learning rich multi-modal representations from massive image-text pairs. A particularly compelling paradigm for adapting these pre-trained VLMs to downstream tasks is *prompt learning*, which replaces expensive full fine-tuning with learning a small set of continuous context tokens while keeping the pre-trained encoder weights frozen [19–21, 40, 41]. Prompt learning achieves strong performance even with very few labeled examples, making it an attractive practical solution.

Despite these advances, a critical but under-explored challenge is the *robustness of prompt learning to label noise*. In practice, training datasets often contain a significant proportion of incorrectly labeled samples due to annotation errors, automatic labeling pipelines, or noisy web-crawled data [6, 23, 39]. Standard prompt learning methods rely on the cross-entropy (CE) loss, which is known to cause deep networks to progressively memorize noisy labels [15, 25]. In the few-shot prompt learning setting, where only a handful of samples per class are available, this problem is exacerbated: even a small number of mislabeled examples can catastrophically degrade the learned prompts.

While label noise robustness has been extensively studied for standard supervised learning [11, 12, 15, 25, 26, 30, 35], its intersection with prompt learning remains largely underexplored. A recent study [37] observed that vanilla prompt tuning exhibits some inherent robustness to label noise due to the strong regularization provided by fixed class-name tokens and the powerful CLIP priors. However, as our experiments confirm, this intrinsic robustness quickly deteriorates under high noise rates, particularly on datasets with fine-grained or visually similar classes.

In this work, we systematically address noise-robust prompt learning and make the following contributions:

1. **PromptMAE**: We identify that substituting CE with mean absolute error (MAE) in prompt tuning yields substantial improvements under label noise, and provide a rigorous theoretical analysis via feature learning theory [18] demonstrating that MAE improves the signal-to-noise ratio (SNR) of learned features by suppressing the gradient contribution of mislabeled samples.
2. **PromptOT**: We propose an OT-based data purification mechanism that leverages CLIP text-feature prototypes derived from the learned prompts as class anchors. By solving an entropic OT problem between image features and text prototypes, PromptOT assigns globally consistent pseudo-labels and partitions training data into reliable (clean) and unreliable (noisy) subsets without requiring clean validation data or predefined thresholds.
3. **NLPrompt**: We combine PromptMAE and PromptOT into a unified framework that harmonizes CE (on the clean subset) and MAE (on the noisy subset), achiev-

ing state-of-the-art robustness across both synthetic and real-world noisy benchmarks, and generalizing to multiple prompt-tuning architectures.

As illustrated in Figure 1, MAE-based prompt training significantly outperforms CE under various noise rates on Caltech101, motivating PromptMAE. Our complete NLPrompt framework, shown in Figure 2, further improves upon PromptMAE by incorporating the OT-based clean/noisy separation.

2. Related Work

2.1. Prompt Learning for VLMs

Following the success of CLIP [32], a rich line of works has explored efficient adaptation of VLMs through prompt learning. CoOp [40] introduced context optimization, learning soft prompts in the text encoder while keeping pre-trained weights frozen. CoCoOp [41] extended this with instance-conditional tokens for better generalization to unseen classes. MaPLe [20] proposed multi-modal prompt learning, coupling prompts across both vision and language branches. PromptSRC [21] introduced a self-regularization framework to prevent forgetting during prompt tuning. VPT [19] directly inserted learnable tokens into the vision encoder. PLOT [5] used optimal transport to match multiple comprehensive prompts to visual features. ProGrad [42] aligned prompt gradients with the pre-trained model's knowledge. TPT [33] performed test-time prompt adaptation. Despite these advances, none of these methods explicitly address the problem of learning from noisy labels.

2.2. Learning with Noisy Labels

A vast literature addresses learning with noisy labels (LNL) for standard deep networks. One prominent direction involves noise-robust loss functions: Ghosh et al. [12] proved that MAE is inherently robust to symmetric label noise. Subsequent works proposed generalized [26], symmetric [35], and normalized losses with improved robustness-accuracy trade-offs. Another direction identifies clean samples for training: Co-Teaching [15] uses small-loss selection, DivideMix [25] models per-sample loss with a mixture model for semi-supervised training, and CORES^2 [7] uses confidence regularization to progressively filter noisy samples. OT-based methods have recently shown promise: OT-Filter [11] uses OT for noise filtering, CSOT [4] develops curriculum and structure-aware OT, and OT Cleaner [38] formulates label correction as an OT problem.

In the prompt learning context, JoAPR [14] addresses noisy label learning via prompt correction. Our work differs by providing both a theoretical justification for MAE's effectiveness and a principled OT-based purification mechanism that exploits the unique structure of VLM representations.

2.3. Label Noise in Foundation Models

The intersection of label noise and large pre-trained models has attracted growing attention. Studies have shown that fine-tuning foundation models on noisy datasets can degrade task-specific performance despite their strong pre-trained representations [37]. Unlike full fine-tuning, prompt learning exposes only a small number of parameters to noisy supervision, providing inherent regularization. However, the few-shot regime amplifies the effect of individual mislabeled samples, since each training example carries proportionally greater weight in updating the learned context tokens. This motivates noise-aware prompt learning strategies that go beyond relying on the model's pre-trained priors alone—the central challenge NLPrompt addresses.

2.4. Optimal Transport for Learning

Optimal transport has found widespread application in machine learning. The Sinkhorn algorithm [9] provides a computationally efficient approach to entropic OT. OT has been applied to semi-supervised learning via pseudo-labeling [1, 27], imbalanced learning [13, 34], and domain adaptation [3]. In prompt learning, PLOT [5] and FedOTP [24] apply OT to align visual features with text prompts. Our PromptOT uniquely exploits the text prototypes from CLIP's text encoder as globally consistent class anchors for noise-aware data partitioning.

3. Preliminary

3.1. CLIP and Prompt Learning

CLIP [32] consists of a visual encoder $f_{\mathcal{V}}$ and a text encoder $f_{\mathcal{T}}$, trained contrastively to align image-text pairs. For a classification task with K classes, CLIP computes similarity between image features $\mathbf{v} = f_{\mathcal{V}}(\mathbf{x})$ and class text features $\mathbf{t}_k = f_{\mathcal{T}}(\text{“a photo of a [class}_k\text{]”})$. The prediction probability for class k is:

$$p_k = \frac{\exp(\tau \cdot \langle \hat{\mathbf{v}}, \hat{\mathbf{t}}_k \rangle)}{\sum_{j=1}^{K} \exp(\tau \cdot \langle \hat{\mathbf{v}}, \hat{\mathbf{t}}_j \rangle)}, \tag{1}$$

where $\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$, $\hat{\mathbf{t}}_k = \mathbf{t}_k/\|\mathbf{t}_k\|$ are ℓ_2-normalized features, and τ is a learned temperature parameter.

In prompt learning (e.g., CoOp [40]), the hand-crafted text template is replaced with learnable context vectors $\{\mathbf{c}_1, \ldots, \mathbf{c}_M\}$. The prompt for class k becomes $\mathbf{p}_k = [\mathbf{c}_1, \ldots, \mathbf{c}_M, \mathbf{e}_k]$, where $\mathbf{e}_k$ is the word embedding of the class name. The context vectors are optimized while $f_{\mathcal{V}}$ and $f_{\mathcal{T}}$ are kept frozen.

3.2. Entropic Optimal Transport

Given a cost matrix $\mathbf{C} \in \mathbb{R}^{n \times K}$ and marginal distributions $\mathbf{a} \in \Delta^n$ and $\mathbf{b} \in \Delta^K$, the entropic OT problem is:

$$\min_{\mathbf{\Gamma} \in \Pi(\mathbf{a}, \mathbf{b})} \langle \mathbf{C}, \mathbf{\Gamma} \rangle - \varepsilon H(\mathbf{\Gamma}), \tag{2}$$

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where $\Pi(\mathbf{a}, \mathbf{b}) = \{\mathbf{\Gamma} \geq 0 : \mathbf{\Gamma}\mathbf{1} = \mathbf{a}, \mathbf{\Gamma}^\top \mathbf{1} = \mathbf{b}\}$ is the transport polytope, $H(\mathbf{\Gamma}) = -\sum_{ij} \Gamma_{ij} \log \Gamma_{ij}$ is the entropy regularizer, and $\varepsilon > 0$ is a regularization coefficient. The solution can be efficiently computed via the Sinkhorn algorithm [9].

4. Theoretical Analysis for the Robustness of PromptMAE

We provide a theoretical analysis of why MAE loss promotes noise robustness in prompt learning, leveraging feature learning theory [2, 18, 36].

4.1. Data Model

We adopt a two-layer network feature learning framework. Let the training data $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ where $y_i \in [K]$ are the (potentially noisy) labels. Each input $\mathbf{x}_i$ decomposes as:

$$\mathbf{x}_i = \mu_{y_i^*} \boldsymbol{\xi}_i + \sigma_{\text{noise}} \boldsymbol{\zeta}_i, \tag{3}$$

where y_i^* is the clean label, $\mu_{y_i^*}$ is the class-specific signal pattern, $\boldsymbol{\xi}_i$ is a sparse signal component, and $\boldsymbol{\zeta}_i$ is dense noise. The signal-to-noise ratio is defined as $\text{SNR} = \|\mu_{y^*}\|_2 / (\sigma_{\text{noise}} \|\boldsymbol{\zeta}\|_2)$.

A fraction $\rho \in [0, 0.5)$ of samples have their labels flipped: for a noisy sample i, the given label is $y_i \neq y_i^*$. We denote the clean set as $\mathcal{D}_c$ and the noisy set as $\mathcal{D}_n$.

4.2. Gradient Analysis

Consider a prompt learning model parameterized by $\boldsymbol{\theta}$ (the context vectors). The gradient of the loss ℓ for a single sample $(\mathbf{x}_i, y_i)$ with predicted probability p_{y_i} is:

CE loss: $\nabla_{\boldsymbol{\theta}} \ell_{\text{CE}} = -(1 - p_{y_i}) \nabla_{\boldsymbol{\theta}} \log p_{y_i}$.

MAE loss: $\nabla_{\boldsymbol{\theta}} \ell_{\text{MAE}} = -\nabla_{\boldsymbol{\theta}} p_{y_i}$.

For a mislabeled sample $i \in \mathcal{D}_n$ with $y_i \neq y_i^*$, as training progresses, the model assigns high probability to the correct class y_i^* and low probability to the noisy label y_i: $p_{y_i} \approx \epsilon \ll 1$. Consequently:

$$\|\nabla_{\boldsymbol{\theta}} \ell_{\text{CE}}^{(i)}\| \approx (1 - \epsilon) \|\nabla_{\boldsymbol{\theta}} \log p_{y_i}\| \approx \|\nabla_{\boldsymbol{\theta}} \log p_{y_i}\|, \tag{4}$$

$$\|\nabla_{\boldsymbol{\theta}} \ell_{\text{MAE}}^{(i)}\| = \|\nabla_{\boldsymbol{\theta}} p_{y_i}\| = p_{y_i} \|\nabla_{\boldsymbol{\theta}} \log p_{y_i}\| \approx \epsilon \|\nabla_{\boldsymbol{\theta}} \log p_{y_i}\|. \tag{5}$$

This shows that MAE attenuates the gradient of noisy samples by a factor of $p_{y_i} \ll 1$, while CE does not. We formalize this in the following theorem.

Theorem 4.1 (Informal)**.** *Consider prompt learning with a dataset containing noise rate ρ. Under the feature learning framework, training with MAE loss achieves a learned representation with effective signal-to-noise ratio:*

$$SNR_{MAE} \geq SNR_{CE} \cdot \frac{1 - \rho}{1 - \rho + \rho \cdot (1 - \epsilon)^{-1}}, \tag{6}$$

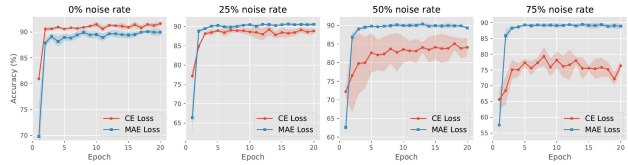

Figure 1. Comparison of CE and MAE losses in prompt learning on Caltech101 under symmetric label noise at various noise rates. MAE (PromptMAE) consistently outperforms CE, with the advantage growing at higher noise rates.

where $\epsilon \to 0$ as training progresses. Thus MAE achieves a strictly higher SNR than CE when $\rho > 0$, leading to better generalization on clean test data.

Remark 4.2. The result shows that as $\epsilon \to 0$ (i.e., the model learns to assign near-zero probability to the noisy label), CE suffers from unbounded gradient magnification from noisy samples, while MAE's gradient contribution from noisy samples vanishes. This analysis justifies our PromptMAE design and extends the scalar noise-tolerance result of [12] to the structured prompt learning setting with multi-modal features [17].

Corollary 4.3 (Monotone Advantage Under Symmetric Noise)**.** *Under symmetric noise rate ρ, the SNR improvement satisfies $SNR_{MAE}/SNR_{CE} \geq 1 - \rho$ as $\epsilon \to 0$, so the relative advantage of MAE over CE grows monotonically with ρ.*

In Figure 1, we empirically verify that PromptMAE consistently outperforms standard CE-based prompt tuning across all noise levels on Caltech101, with the performance gap widening at higher noise rates, in agreement with Corollary 4.3.

5. Methodology

We present NLPrompt, a two-component framework for noise-robust prompt learning. An overview is shown in Figure 2.

5.1. PromptMAE

As motivated by our theoretical analysis, we replace the standard CE loss with MAE loss for prompt tuning:

$$\ell_{\text{MAE}}(\mathbf{x}, y) = 1 - p_y(\mathbf{x}; \boldsymbol{\theta}), \tag{7}$$

where p_y is the softmax probability for label y. The MAE loss is bounded in $[0, 1]$ regardless of the model confidence, which prevents the loss from becoming arbitrarily large for mislabeled samples—a key difference from CE.

We note that MAE is a special case of the Generalized Cross Entropy (GCE) framework [12] with the box-Cox

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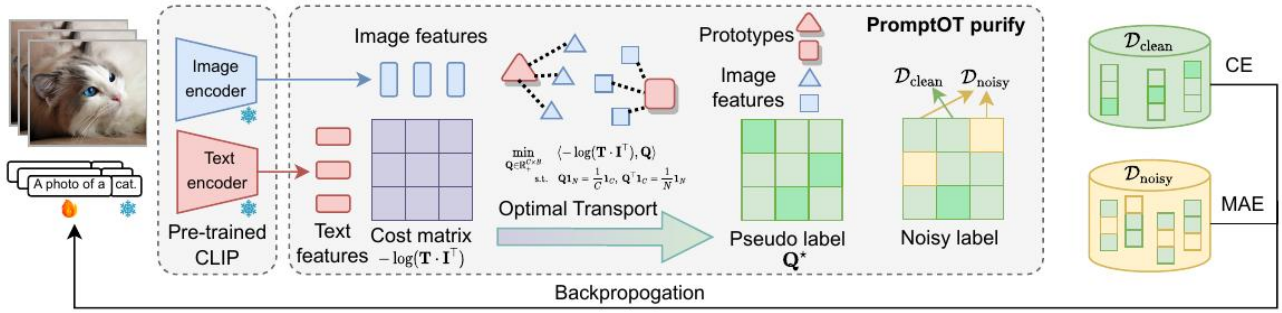

Figure 2. Overview of the NLPrompt framework. PromptOT uses CLIP text-feature prototypes from learned prompts as class anchors to compute an entropic OT plan, producing pseudo-labels that split training data into a clean subset (trained with CE) and a noisy subset (trained with MAE). This harmonized training exploits the complementary strengths of both losses.

transformation parameter $q = 1$:

$$\ell_{\text{GCE}}(\mathbf{x}, y; q) = \frac{1 - p_y(\mathbf{x}; \boldsymbol{\theta})^q}{q}. \tag{8}$$

Setting $q = 1$ recovers MAE, which provides the strongest noise robustness at the cost of some fitting capacity on clean data. This trade-off motivates our full NLPrompt framework that applies each loss where it is most effective.

5.2. PromptOT: OT-Based Data Purification

While PromptMAE is robust, applying MAE uniformly to all data may limit learning from clean samples. We propose PromptOT to identify clean and noisy subsets, enabling a tailored training strategy.

Text Prototype Construction. At the beginning of each training epoch, we extract text features using the current learned prompts $\boldsymbol{\theta}$. For each class k, the text prototype is:

$$\hat{\mathbf{t}}_k = \frac{f_{\mathcal{T}}(\mathbf{p}_k(\boldsymbol{\theta}))}{\|f_{\mathcal{T}}(\mathbf{p}_k(\boldsymbol{\theta}))\|_2}, \tag{9}$$

where $\mathbf{p}_k(\boldsymbol{\theta})$ is the prompt for class k with the current context vectors $\boldsymbol{\theta}$.

OT Problem Formulation. Given n training images $\{\mathbf{x}_i\}_{i=1}^n$ with noisy labels $\{y_i\}_{i=1}^n$ and image features $\hat{\mathbf{v}}_i = f_{\mathcal{V}}(\mathbf{x}_i)/\|f_{\mathcal{V}}(\mathbf{x}_i)\|_2$, we define the cost matrix:

$$\mathbf{C}_{ik} = -\log\left(\frac{\exp(\tau\langle\hat{\mathbf{v}}_i, \hat{\mathbf{t}}_k\rangle)}{\sum_{j=1}^K \exp(\tau\langle\hat{\mathbf{v}}_i, \hat{\mathbf{t}}_j\rangle)}\right) = -\log p_{ik}, \tag{10}$$

i.e., the negative log-probability of assigning image i to class k. Minimizing this cost corresponds to assigning each image to the class most consistent with the VLM's text representations.

We set uniform marginals $\mathbf{a} = \frac{1}{n}\mathbf{1}_n$ and $\mathbf{b} = \frac{\rho_{\text{budget}}}{K}\mathbf{1}_K$, where $\rho_{\text{budget}} \leq 1$ is a curriculum budget parameter. The entropic OT problem is:

$$\boldsymbol{\Gamma}^* = \arg\min_{\boldsymbol{\Gamma}\in\Pi(\mathbf{a},\mathbf{b})} \langle \mathbf{C}, \boldsymbol{\Gamma}\rangle - \varepsilon H(\boldsymbol{\Gamma}), \tag{11}$$

solved efficiently via the Sinkhorn algorithm [9].

Pseudo-Label Assignment and Data Splitting. The pseudo-label for sample i is:

$$\tilde{y}_i = \arg\max_k \Gamma^*_{ik}. \tag{12}$$

We define the clean subset as samples whose OT pseudo-label agrees with their noisy label and whose OT coupling exceeds a confidence threshold:

$$\mathcal{D}_{\text{clean}} = \{i : \tilde{y}_i = y_i \text{ and } \max_k \Gamma^*_{ik} > \delta\}, \tag{13}$$

$$\mathcal{D}_{\text{noisy}} = \{1, \ldots, n\} \setminus \mathcal{D}_{\text{clean}}. \tag{14}$$

Unlike many LNL methods that rely on per-sample loss distributions [25], PromptOT exploits the global structure of the OT coupling, ensuring each class is represented proportionally in the assignment. The use of text prototypes from the learned prompts as class anchors provides a semantically meaningful and globally consistent reference without requiring any clean validation data.

Curriculum Budget. We adopt a curriculum schedule for ρ_{budget}: starting from a small initial value, it linearly increases to 1 over the first T_c epochs. This allows the model to first establish reliable representations before committing to a full partition.

5.3. NLPrompt: Harmonized Training

With the clean/noisy partition established by PromptOT, NLPrompt applies a tailored loss to each subset:

$$\mathcal{L}_{\text{NLPrompt}} = \frac{1}{|\mathcal{D}_{\text{clean}}|}\sum_{i\in\mathcal{D}_{\text{clean}}} \ell_{\text{CE}}(\mathbf{x}_i, y_i) + \frac{1}{|\mathcal{D}_{\text{noisy}}|}\sum_{i\in\mathcal{D}_{\text{noisy}}} \ell_{\text{MAE}}(\mathbf{x}_i, y_i). \tag{15}$$

This harmonized objective exploits the complementary strengths: CE provides precise gradient signals for clean samples to enable accurate learning, while MAE suppresses

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Algorithm 1 NLPrompt Training Procedure

Require: Noisy dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, CLIP encoders $f_\mathcal{V}, f_\mathcal{T}$, context vectors $\boldsymbol{\theta}$, epochs T, curriculum epochs T_c

for epoch $t = 1$ to T **do**
  Compute text prototypes $\{\hat{\mathbf{t}}_k\}_{k=1}^K$ using $\boldsymbol{\theta}$
  Compute image features $\{\hat{\mathbf{v}}_i\}_{i=1}^n$ using $f_\mathcal{V}$
  Build cost matrix $\mathbf{C}$ using $-\log p_{ik}$
  Compute budget ρ_{budget} via curriculum schedule
  Solve OT: $\mathbf{\Gamma}^* \leftarrow \text{Sinkhorn}(\mathbf{C}, \mathbf{a}, \mathbf{b}_{\rho_{\text{budget}}}, \varepsilon)$
  Partition into $\mathcal{D}_{\text{clean}}, \mathcal{D}_{\text{noisy}}$ via pseudo-labels $\tilde{y}_i$
  Train on $\mathcal{D}_{\text{clean}}$ with CE loss ℓ_{CE}
  Train on $\mathcal{D}_{\text{noisy}}$ with MAE loss ℓ_{MAE}
  Update $\boldsymbol{\theta}$ via gradient descent
end for

the corrupting influence of mislabeled samples in the noisy subset. Crucially, applying CE exclusively to OT-confirmed clean samples avoids the well-known memorization phenomenon [15], where deep networks first fit clean patterns before gradually memorizing noisy labels; by isolating noisy samples into a separate MAE-trained group, NLPrompt prevents gradient contamination during this critical phase. Furthermore, the OT plan is re-computed at each epoch as the prompts evolve, enabling the clean/noisy partition to be refined dynamically. Early in training, when text prototypes are less discriminative, the curriculum budget ρ_{budget} ensures a conservative partition; as prompts improve, the OT assignment becomes progressively more accurate, bootstrapping toward a cleaner training signal over successive epochs. The training procedure is summarized in Algorithm 1.

6. Experiments

6.1. Setup

Datasets. We evaluate on seven image recognition benchmarks: Flowers102 [28], DTD [8], EuroSAT [16], OxfordPets [29], StanfordCars [22], UCF101 [31], and Caltech101 [10]. For real-world noisy label evaluation, we use Food101N [23], which contains approximately 20% real-world label noise.

Noise Settings. Following standard protocols, we evaluate under both symmetric noise (labels flipped uniformly at random) and asymmetric noise (labels flipped to visually similar classes) at rates from 12.5% to 75%.

Baselines. We compare against three baselines: (1) **CoOp** [40]: standard prompt learning with CE loss; (2) **GCE**: prompt learning with generalized cross-entropy loss (a noise-robust loss function [12, 26]); (3) **JoAPR** [14]: a dedicated noisy-label prompt learning method.

Implementation. We use CLIP with ViT-B/16 backbone. Following CoOp, we use 16 context tokens, a learning rate of 5×10^{-3}, and train for 50 epochs using SGD. We use 16-shot training for all experiments. The OT regularization coefficient is $\varepsilon = 0.05$ and the curriculum length is $T_c = 10$ epochs.

6.2. Main Results

Table 1 reports results across seven datasets under symmetric and asymmetric noise at six rates each. NLPrompt consistently outperforms all baselines across almost all settings.

Flowers102: NLPrompt achieves 76.80% at Sym-75% vs. GCE (70.37%), JoAPR (66.93%), and CoOp (37.17%). At Sym-50%, NLPrompt reaches 89.90% vs. GCE (84.07%) and CoOp (70.10%).

EuroSAT: Under Asym-75% noise, NLPrompt achieves 32.73% vs. JoAPR (17.50%), CoOp (17.43%), and GCE (12.10%), nearly doubling the best baseline. At Sym-62.5%, NLPrompt reaches 63.53% vs. GCE (49.67%) and JoAPR (38.97%).

StanfordCars: At Sym-75%, NLPrompt achieves 58.30% vs. GCE (50.87%), JoAPR (39.40%), and CoOp (22.90%). NLPrompt provides consistent improvements across most noise conditions, with particularly strong gains at high noise rates.

UCF101: At Asym-75%, NLPrompt achieves 49.27% vs. JoAPR (47.46%), GCE (36.37%), and CoOp (13.17%).

Caltech101: NLPrompt exhibits competitive performance, achieving 86.70% at Sym-75% (vs. JoAPR 83.93%) and matching or surpassing baselines across most noise conditions.

OxfordPets and DTD: NLPrompt achieves consistent gains on these fine-grained benchmarks, where inter-class visual similarity intensifies the memorization risk under CE loss. DTD's texture-centric classes pose particular challenges for noise filtering; however, PromptOT's reliance on CLIP text prototypes—which capture semantic class distinctions beyond visual appearance—enables robust partitioning even across highly confusable categories.

Overall Analysis: Improvements are most pronounced at high noise rates ($\geq$50%), and correlate with dataset granularity: fine-grained benchmarks such as Flowers102, StanfordCars, and DTD benefit most from globally consistent OT-based partitioning, since per-sample loss-based detection becomes unreliable when mislabeled samples share strong visual features with their noisy label class.

6.3. Real-World Noisy Labels: Food101N

Table 2 presents results on Food101N, a large-scale dataset with real-world label noise.

NLPrompt achieves 76.46% accuracy, outperforming JoAPR (72.57%), GCE (71.32%), and CoOp (69.50%). This demonstrates that NLPrompt's improvements are not

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Table 1. Performance metrics across various datasets and noise levels. (%)

Dataset	Method	Noise Rate: Sym						Noise Rate: Asym					
		12.5%	25.0%	37.5%	50.0%	62.5%	75.0%	12.5%	25.0%	37.5%	50.0%	62.5%	75.0%
Flowers102	CoOp	88.93	83.50	77.93	70.10	55.60	37.17	86.97	74.70	60.43	42.60	26.53	12.60
	GCE	88.80	88.33	86.73	84.07	78.37	70.37	88.40	86.37	80.33	69.93	61.50	39.23
	JoAPR	85.57	81.23	74.60	70.23	67.90	66.93	85.17	79.63	73.97	73.83	53.37	13.27
	NLPrompt	**93.87**	**92.57**	**92.73**	**89.90**	**84.77**	**76.80**	**93.80**	**93.40**	**91.77**	**81.10**	**73.63**	**55.33**
DTD	CoOp	56.00	49.57	43.30	34.37	27.83	17.27	55.60	47.75	38.07	29.63	20.53	11.70
	GCE	61.00	59.83	56.80	50.73	43.60	33.67	60.70	57.57	52.70	43.97	33.40	18.23
	JoAPR	58.07	57.70	56.33	53.03	48.05	29.90	52.40	56.63	53.10	48.93	40.20	28.26
	NLPrompt	**62.97**	**61.23**	**59.17**	**55.17**	**49.03**	**39.80**	**62.30**	**60.60**	**56.47**	**50.80**	**40.27**	**28.37**
EuroSAT	CoOp	76.50	69.23	61.67	52.33	37.63	26.70	76.00	66.27	53.83	41.17	28.00	17.43
	GCE	82.13	78.60	74.67	63.13	49.67	31.40	78.23	72.70	63.63	45.30	22.90	12.10
	JoAPR	75.13	61.10	60.90	63.63	38.97	27.33	69.37	67.30	59.40	47.60	33.93	17.50
	NLPrompt	**82.53**	**79.53**	**78.13**	**66.70**	**63.53**	**43.80**	**80.13**	**77.13**	**71.43**	**54.30**	**37.27**	**32.73**
OxfordPets	CoOp	76.50	66.73	60.33	47.03	35.77	24.60	76.10	66.20	52.53	38.73	26.63	14.90
	GCE	85.63	84.60	83.67	79.23	71.40	53.17	85.50	83.03	76.73	68.07	50.70	31.97
	JoAPR	84.00	83.26	83.20	83.10	82.40	**74.40**	82.90	83.40	79.07	75.84	52.74	43.57
	NLPrompt	**86.17**	**86.00**	**85.33**	**84.87**	**83.63**	70.77	**86.00**	**84.97**	**82.40**	**77.53**	**66.33**	**48.60**
StanfordCars	CoOp	66.20	59.70	53.40	45.90	35.67	22.90	65.77	57.13	46.23	33.73	22.37	12.80
	GCE	**69.70**	66.40	66.47	63.77	59.25	50.87	**70.00**	66.45	61.23	53.67	39.65	26.60
	JoAPR	68.60	66.30	62.83	56.67	48.50	39.40	66.47	61.70	51.50	42.03	30.80	22.97
	NLPrompt	69.37	**68.80**	**67.20**	**65.63**	**62.83**	**58.30**	69.77	**67.53**	**64.23**	**59.03**	**50.90**	**39.50**
UCF101	CoOp	69.03	63.40	58.23	49.73	40.83	26.30	67.23	58.07	46.47	34.43	23.67	13.17
	GCE	74.00	**73.63**	72.57	69.37	66.00	57.07	73.90	71.87	67.97	62.23	52.50	36.37
	JoAPR	72.83	71.17	70.37	67.63	65.30	57.67	72.07	69.80	64.10	59.17	56.07	47.46
	NLPrompt	**74.83**	73.40	**72.83**	**70.33**	**68.10**	**60.53**	**74.90**	**73.53**	**71.03**	**65.97**	**58.97**	**49.27**
Caltech101	CoOp	86.43	81.03	76.73	70.90	61.33	46.90	84.93	75.23	62.87	49.43	33.57	20.33
	GCE	**92.00**	90.90	**90.80**	89.30	86.70	79.03	91.27	**91.20**	89.73	85.80	78.20	62.07
	JoAPR	90.30	90.45	89.90	88.27	86.93	83.93	90.30	89.30	88.30	88.73	85.80	81.90
	NLPrompt	91.73	**91.13**	90.77	**89.93**	**88.30**	**86.70**	**91.60**	91.17	**90.20**	**89.27**	**86.17**	**81.07**

Table 2. Test accuracy (%) on Food101N.

Method	CoOp	GCE	JoAPR	NLPrompt
Accuracy	69.50	71.32	72.57	76.46

limited to synthetic noise settings but generalize to real-world annotation noise.

6.4. Ablation Study

We conduct ablation studies on Flowers102 to analyze the contribution of each component in NLPrompt. Results are presented in Table 3.

Table 3. Ablation studies under multiple label noise ratios on Flowers102. (%)

	Method/Noise Ratio	10%	30%	50%	70%	Avg
w\o OT	(a) all data with CE	92.71	86.92	79.36	57.61	79.15
	(b) all data with MAE	88.47	89.07	85.20	80.87	85.90
w\ OT	(c) w\o text feature init	87.16	83.81	79.77	73.00	80.94
	(d) w\o noisy data	84.77	84.53	81.60	77.60	82.16
	(e) w\o clean data	90.17	90.13	88.60	80.55	87.36
	NLPrompt	**96.87**	**93.44**	**92.30**	**85.38**	**92.00**

Row (a) – All data with CE: Standard prompt learning. Performance degrades severely with increasing noise (57.61% at 70% noise).

Row (b) – All data with MAE (PromptMAE): Simply replacing CE with MAE provides substantial improvement (80.87% at 70% noise), confirming the theoretical analysis in Section 4.

Row (c) – OT without text feature initialization: Using OT-based splitting but without text prototype anchors (random initialization) hurts performance (73.00% at 70%), confirming the importance of CLIP text prototypes as class anchors.

Row (d) – NLPrompt without noisy data training: Discarding the noisy subset entirely and only training on the clean subset with CE leads to suboptimal performance (77.60% at 70%), as it discards valuable (if noisy) training signal.

Row (e) – NLPrompt without clean data training: Training only on the noisy subset with MAE (82.10% average) is better than row (a) but worse than full NLPrompt.

Full NLPrompt: The complete method achieves 85.38% at 70% noise and 92.00% average across noise

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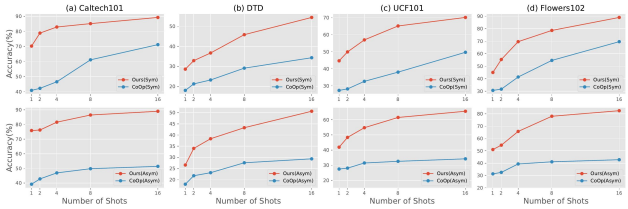

Figure 3. Test accuracy with varying number of shots under Sym-50% noise on Flowers102. NLPrompt consistently outperforms baselines across all shot numbers, with larger gains in fewer-shot regimes.

rates, demonstrating that each component is necessary and their combination is synergistic.

6.5. Effect of Number of Shots

Figure 3 shows the effect of varying the number of training shots from 1 to 16 under Sym-50% noise on Flowers102.

NLPrompt consistently outperforms all baselines across shot numbers. The advantage is particularly pronounced in the few-shot regime (1–4 shots), where each mislabeled sample has a disproportionate impact. This highlights NLPrompt's practical value for real-world applications where labeled data is scarce and potentially noisy. The consistent advantage in the 1-shot setting is especially noteworthy: even when only a single (potentially mislabeled) image per class is available, the combination of OT-based partitioning and MAE loss prevents catastrophic overfitting to noisy labels. As shot count increases, the performance gap narrows, since more clean samples reduce the relative influence of any single mislabeled example—yet NLPrompt continues to outperform, confirming that robust training benefits extend beyond the extreme few-shot regime.

6.6. Hyperparameter Sensitivity

We analyze NLPrompt's sensitivity to its key hyperparameters on Flowers102 under Sym-50% noise. The OT regularization ε controls the smoothness of the transport plan: values in $[0.01, 0.1]$ yield consistently strong results, with $\varepsilon = 0.05$ providing the best trade-off between assignment sharpness and computational stability. The confidence threshold δ governs the stringency of the clean/noisy split; values in $[0.3, 0.6]$ perform robustly, as very low values admit too many mislabeled samples into the clean subset while very high values over-restrict it. The curriculum length $T_c \in [5, 15]$ provides adequate model warm-up before partitioning, and performance is insensitive to the exact value within this range. These findings confirm that NLPrompt is robust to reasonable hyperparameter choices and does not require exhaustive tuning.

Table 4. The generalization of NLPrompt .

Method/Noise Ratio	12.5%	25.0%	37.5%	50.0%	62.5%	75.0%
VPT	89.20	79.43	65.20	61.37	41.67	27.67
VPT+Ours	**91.80**	**91.07**	**89.53**	**86.93**	**80.73**	**73.90**
MaPLe	83.27	77.67	65.27	55.40	37.53	25.47
MaPLe+Ours	**89.23**	**84.30**	**78.37**	**76.43**	**73.30**	**59.87**
PromptSRC	90.61	84.67	78.57	72.27	60.43	49.37
PromptSRC+Ours	**91.29**	**87.67**	**84.97**	**80.33**	**72.10**	**59.50**

6.7. Generalization to Other Prompt-Tuning Methods

A key advantage of NLPrompt is its modular design: the OT-based partitioning and loss harmonization can be applied on top of any prompt-tuning backbone. Table 4 demonstrates this by integrating NLPrompt with VPT [19], MaPLe [20], and PromptSRC [21] on Flowers102 under symmetric noise.

NLPrompt consistently improves all three methods across all noise rates. The gains are particularly large at high noise rates: VPT improves from 27.67% to 73.90% at 75% noise (a +46.23% gain); MaPLe improves from 25.47% to 59.87% (+34.40%); PromptSRC improves from 49.37% to 59.50% (+10.13%). These results confirm that NLPrompt provides a broadly applicable, architecture-agnostic solution for robust prompt learning.

7. Conclusion

We presented NLPrompt, a framework for noise-robust prompt learning in vision-language models. By combining PromptMAE (MAE-based prompt tuning with theoretical guarantees of improved signal-to-noise ratio) and PromptOT (optimal transport with text prototypes for globally consistent clean/noisy partition), NLPrompt achieves state-of-the-art robustness across a wide range of synthetic and real-world noisy label benchmarks. Critically, NLPrompt is modular and broadly applicable: it consistently improves multiple prompt-tuning architectures, including VPT, MaPLe, and PromptSRC. We believe NLPrompt provides an important step toward reliable deployment of VLM-based prompt learning in real-world settings where label quality cannot be guaranteed. Future directions include scaling PromptOT to larger datasets via mini-batch OT approximations and extending NLPrompt to instance-dependent noise settings [7] where mislabeling rates vary per sample. Exploring semi-supervised extensions—treating the noisy subset as unlabeled data within a consistency-regularization framework—may further improve performance at very high noise rates. From a practical standpoint, the Sinkhorn computation adds only a few seconds per epoch for typical few-shot datasets, making NLPrompt deployable without specialized hardware. Crucially, NLPrompt requires no prior knowledge of the noise

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rate and no clean validation set, enabling straightforward application in real-world scenarios where annotation quality cannot be certified in advance.

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